TALHA ASHRAF

COMPUTER SYSTEMS ENGINEER

CONTACT

talhanz1112@gmail.com

https://tashraf.netlify.app

Auckland, New Zealand

KEY SKILLS

- Free RTOS
- Safety Critical Programming
- FPGA Development
- Data Analysis
- Python Data Visualisation
- Software Testing

EDUCATION

Bachelor of Engineering (Hons)

University of Auckland

2022 - 2025

Specialising in Computer Systems Engineering.

National Certificate of Educational Achievement (NCEA)

Papatoetoe High School

2017- 2021

Completed level 3 with excellence endorsement.

INTERESTS

- Photography
- 3D printing
- Reading

LANGUAGES

English

Urdu

OBJECTIVE

A **passionate** and **driven** final-year Computer Systems Engineering student with a strong interest in **Embedded Systems,** including **firmware** and **hardware**. Seeking a graduate/intern role and eager to work with a **dynamic** team, **solve interesting problems** and and contribute to making a **positive impact** on the **world**.

EXPERIENCE

Hardware Engineering R&D Intern

Teletrac Navman — Nov 2024 - Feb 2025

- Developed automated device testing workflows using **Python** and the **VISA** communication standard with benchtop **instrumentation.**
- **Synchronized data** across time domains by converting timestamps from various formats into UNIX time.
- Designed a user-friendly and **intuitive UI** to display test data and facilitate the extraction of results from devices under test.
- Diagnosed **RMA** devices by intercepting **RS232** serial communication signals and **simulating interactions** from central hub device, ensuring checksum integrity and adhering to proprietary protocols.

Product Development Intern

Fisher & Paykel Healthcare — Nov 2023 - Feb 2024

- Enhanced user experience of **FPH 950** heated humidifier device by using **firmware** development skills in **C/C++** to optimize usability based on user feedback.
- Ensured code safety using **unit testing** and **integration testing**.
- Conducted automated **development testing** using **C#** in controlled physical environments.
- Documented outcomes to **ensure quality** standards.
- Used **MATLAB** for data processing, analysis, and visualization.
- Provided **insights** to support decision-making and **enhance** project **outcomes**.
- Carried out rigorous **compliance testing** to ensure heaterbase complied with **ISO 80601-2-12** (international standard for medical electrical equipment).

PROJECTS

Wireless Energy Monitor – EVolocity

Project Lead – Hardware, Firmware & Software Integration

- **Led** a **multidisciplinary** team to develop a wireless energy monitoring system for electric carts in national EVolocity competitions.
- **Designed** custom sensing hardware and **PCBs** in **Altium** to accurately capture voltage and current metrics with **<1% accuracy** in challenging mobile environments.
- Programmed firmware in **MicroPython** for the **Raspberry Pi Pico W**, utilizing its dual-core processor, ADC, and onboard storage to log energy data throughout the race.
- Enabled **wireless** post-race data transmission to a polished user interface, giving teams clear, **actionable insights** into energy performance.
- Delivered a **cost-effective, reliable**, and **extensible** system designed to support future competition use and STEM engagement.

Line-Following Robot - PSoC 5LP ARM SoC

Integration & Testing Lead – Systems Validation & Firmware

- Built an **autonomous maze-navigating robot** using a **PSoC 5LP ARM SoC,** onboard sensors, and rotary encoders.
- **Designed** analog signal path using **oscilloscope** and **FFT analysis** to optimize noise rejection.
- Developed **PCB** in **Altium** with **optimized** IR sensor placement for **high-reliability** line detection.
- Implemented **maze-solving** logic and **PI control** in **C,** using encoder feedback and **PWM**-driven motors for smooth motion.
- Led **extensive validation and edge-case testing** across varying maze designs and lighting conditions to ensure **robust, reliable performance.**